Computational Statistical Physics
Exercise 1

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1 Description

This report aims to draw qualitative conclusions regarding the 2D Icing Model, using the Metropolis-Hastings algorithm to simulate spin flips. Of primary interest is deciding the critical temperature T_C for a given lattice of sufficiently large size. The approach will assume no prior knowledge of the exact solution for the 2D model, although it will be used as a reference in some plots throughout the report to make sure the initial simulations are on the right track. For the critical temperature, we refer to Onsager's exact solution in 2D [1]:

$$\frac{k_B T_C}{J} = \frac{2}{\ln(1+\sqrt{2})} \approx 2.269 \tag{1}$$

Here on after in the report, T_C will refer to the exact solution of the critical temperature, while the approximation aimed to be decided will be denoted T_C^*.

2 Lattice Plots

As a first procedure, a 50×50 lattice is simulated for temperatures $T \in \{2, T_C, 2.5\}$ and it's spins plotted. This to give an idea that the simulation works as intended, with a phase transition at T_C. The lattice is first initialized with every spin pointing up, where after 2000 sweeps allow the lattice to equilibrate to it's given temperature, each sweep accounting for at most $L \times L$ possible spin flips.

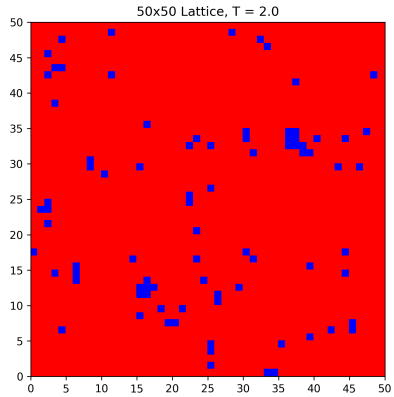

Figure 1: Simulated lattice below critical temperature.

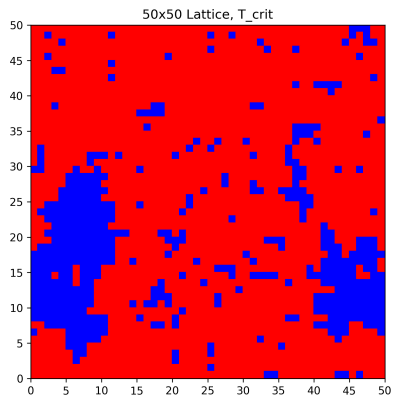

Figure 2: Simulated lattice at critical temperature.

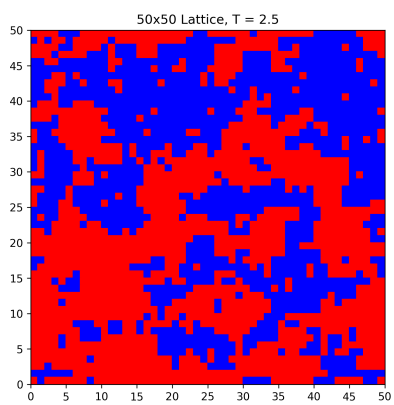

Figure 3: Simulated lattice above critical temperature.

Based on the three plots above, it is evident that a phase transition occurs at T_C. This results in a lowering of the magnetization M and an increase in the energy E, which shall be investigated in the next section.

3 Observables

The variables of interest that can be extracted from the lattice are the energy E, the magnetization M as well as heat capacity C_V and magnetic susceptibility χ given through the variance of the prior variables, according to the fluctuation-dissipation theorem. Lattices of sizes 10×10, 20×20, 50×50 & 100×100 are simulated and plotted correspondingly. The temperature is varied from 1.0 to 3.0 with a step size 0.1. When considering the energy and magnetization, it is clear that their graphs become more smooth for greater lattice sizes. It is also evident that the magnetization has a steeper and more pronounced decline near T_C, for larger lattice sizes. Based on the observables plot, to further estimate T_C^* a lattice size of 50×50 is chosen. This to ensure that the measured observables are reliable, yet the simulations run sufficiently fast.

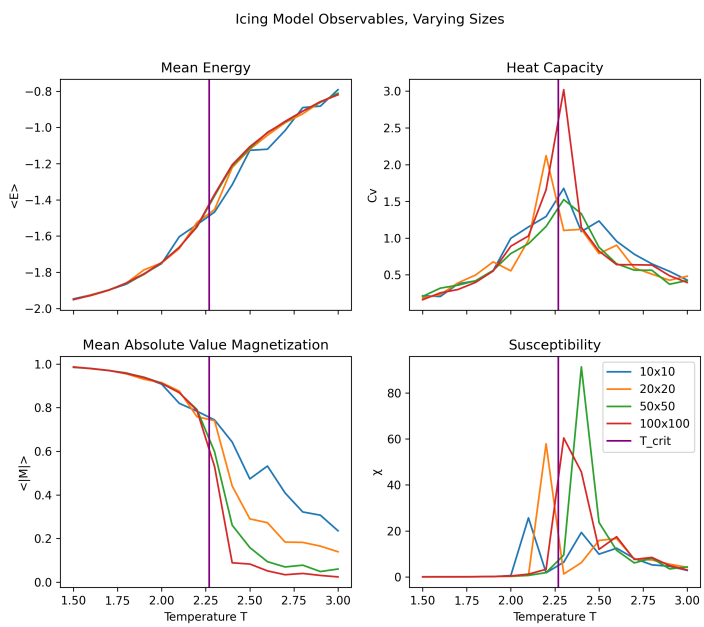

Figure 4: Observables for different lattice sizes. $\langle E \rangle$ and C_V have been normalized by the corresponding number of lattice sites to generate consistent plots.

4 Estimating the Critical Temperature

The approach of estimating T_C^* will consist of running multiple simulations of a 50×50 lattice in a set temperature range. Mainly, the aim is to take the temperature at the point of the maximum value for given response function. This procedure will be performed for the heat capacity, as well as the susceptibility, to decide which response function seems to yield a more valid estimate of T_C^*. A temperature range [2.15, 2.40] with a step size 0.01 is chosen for simulating 50 individual lattices in parallel. T_C^* is then decided by taking the point of maximum temperature for every simulation and storing this temperature in a histogram. Finally, a gaussian curve is fitted to the histogram, which allows for interpreting the gaussian mean value and standard deviation as the critical temperature estimation as well as it's error.

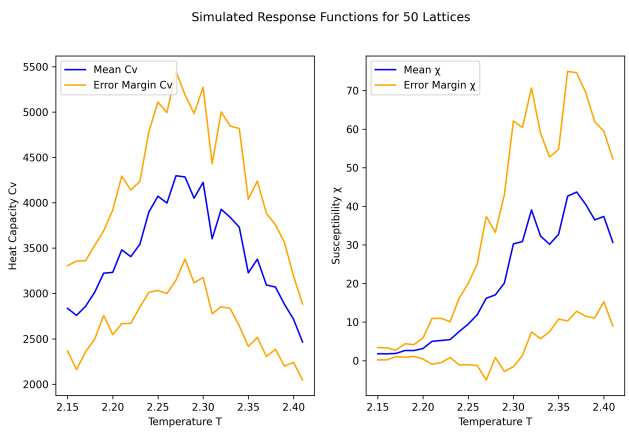

Figure 5: Average C_V and χ for 50 lattice simulations.

Based on the response functions, critical temperature ranges $T_C^* \in [2.257, 2.329]$ and $T_C^* \in [2.314, 2.386]$ can be established for the 50×50 lattice. For this lattice size, given that the simulation outcomes are reasonable, it can be concluded that considering C_V yields a better estimate for the critical temperature. It seems for lattice size 50×50, the susceptibility χ reacts slower as the temperature is increased, thereby a higher critical temperature as compared to the exact answer. This is in alignment with the Susceptibility plot in Figure 4.

References

[1] *Square lattice Ising model — Wikipedia*. Accessed: 2025-03-11. 2024. URL: `https://en.wikipedia.org/wiki/Square_lattice_Ising_model`.

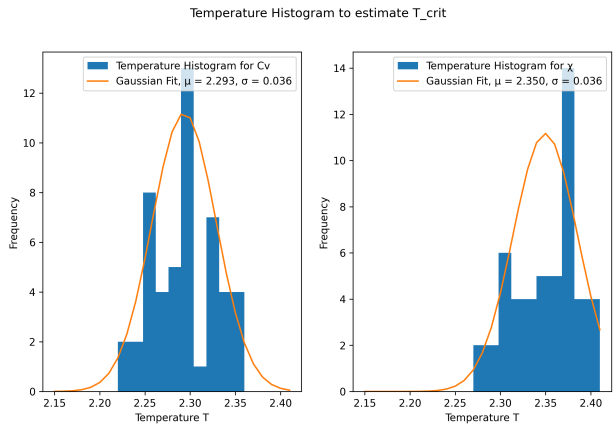

Figure 6: C_V: $T_C^* = 2.293 \pm 0.036$, χ: $T_C^* = 2.350 \pm 0.036$